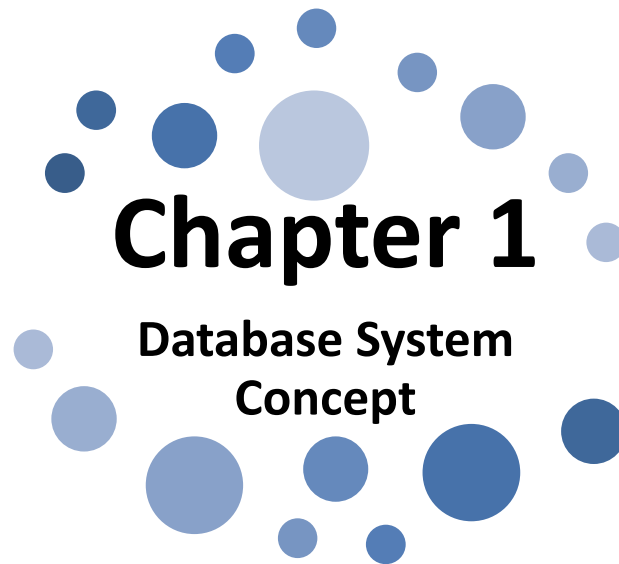




Chapter 1

Database System Concept

Unit 1.6: Data Modelling Using the E-R Model: Entity Relationship Model, Strong Entity Set, Weak Entity Set, Types of Attributes, E-R Diagrams.

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Learning Outcomes

The Learners will be able to:

1. **Draw** the E-R diagram of the given database and **Identify** relationship between entities.

Entity Relationship Model

- **Entity relationship model** defines the conceptual view of database.
- Graphical representation of entities and their relationships in a database structure
- Introduced by Chen in 1976
- Widely accepted and adapted **graphical tool** for data modeling.
- At view level, ER model is considered well for designing databases
- **Entity relationship diagram (ERD)**
 - Uses graphic representations to model database components
 - Each entity in the ER diagram corresponds to a table in the database.
 - The attributes of an entity correspond to fields of a tables
 - ER diagram is converted to the database

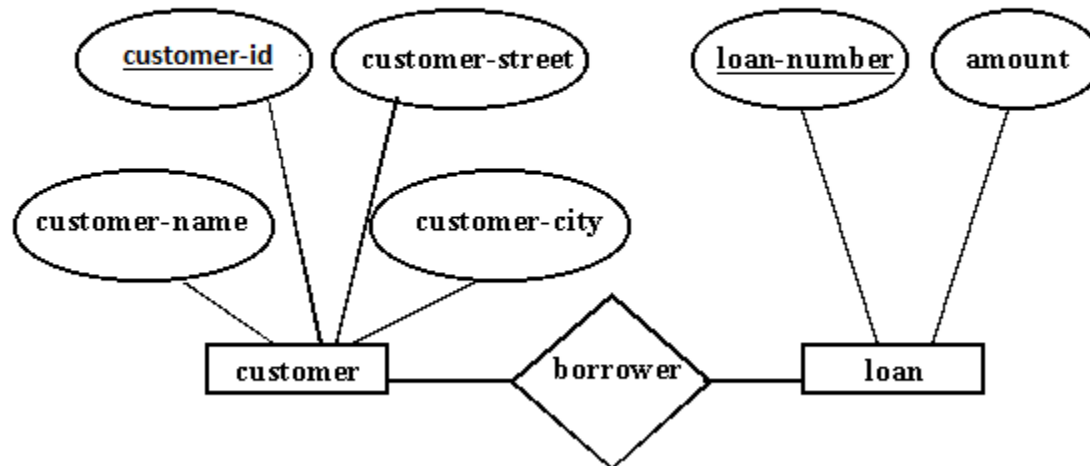
Entity

- **Entity** is a real-world thing that can be easily identifiable and distinguishable.
- An **Entity** can be any object, place, person or class.
- For example, in a school database, student, teachers, class and course offered can be considered as entities.
- All entities have some **attributes** or properties that give them their identity.
- An **entity set** is a collection of similar types of entities.
- An **entity set** is a **group of entities** that possess the **same set of attributes**.
- For example, **Students entity set** may contain all the students of a school; likewise **Teachers entity set** may contain all the teachers of school from all faculties.

Strong Entity Set

- An entity set that has a primary key is called as **Strong entity set**.
- By using a **primary key**, entities in the table can be uniquely identified.
- In an ER diagram, the strong entity set is represented by the **rectangle**. Here, the **primary key** is underlined with the **solid line**.
- A member of a strong entity set is called dominant entity

STRONG ENTITY SET

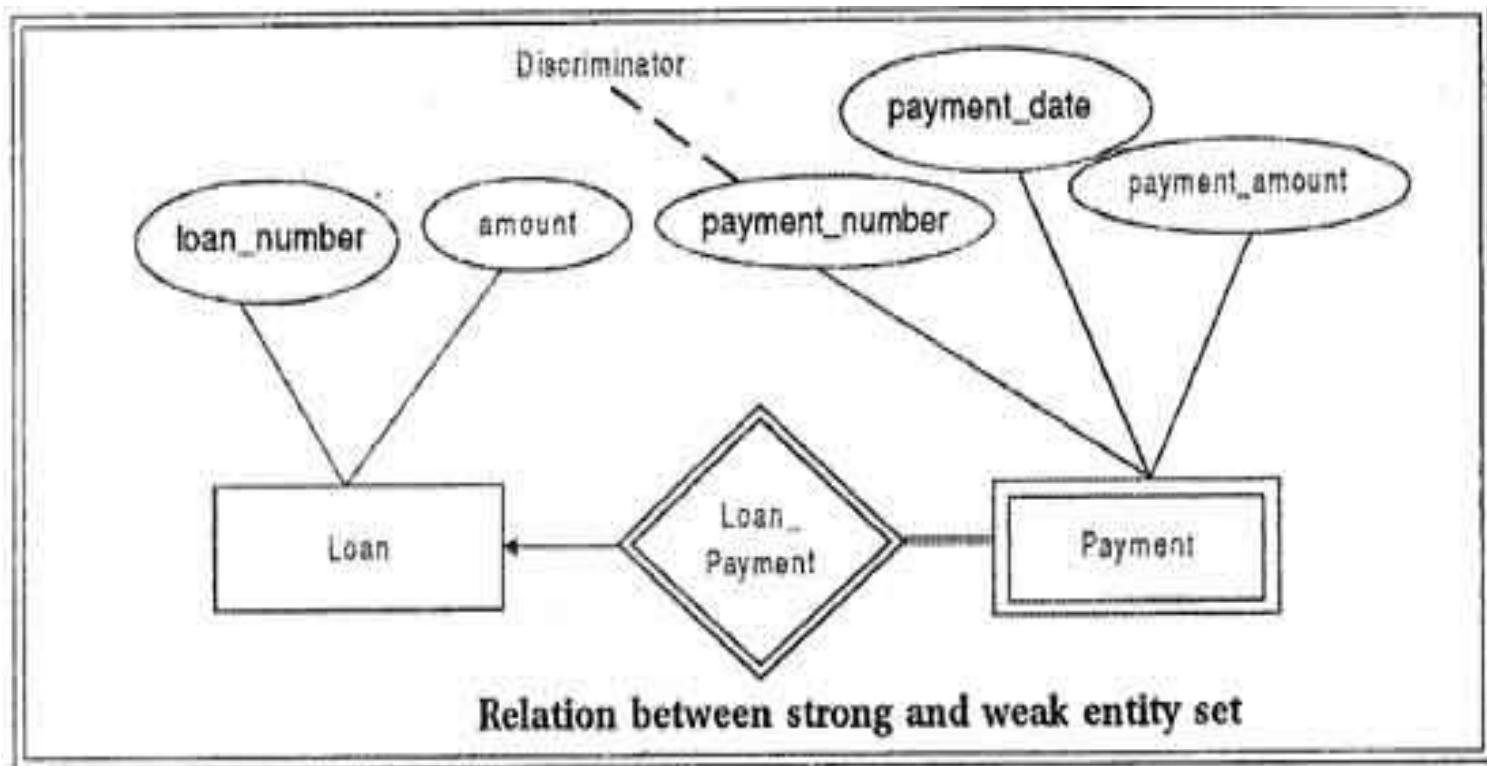


Weak Entity Set

- The entity set which does not have sufficient attributes to form a **primary key** is called as **Weak entity set**.
- But, for discriminating the entities in a set, the weak entity set is **dependent** on a particular **strong entity set**.
- It has to be dependent on **identifying entity set** i.e. a particular 'strong entity set'.
- Double rectangle represents weak entity.
- The relation between a weak entity set and a strong entity set is called as **identifying relationship**.



Weak Entity Set



Strong and Weak Entity Set

Strong Entity Set	Weak Entity Set
It has its own primary key.	It does not have sufficient attributes to form a primary key on its own.
It is represented by a rectangle.	It is represented by a double rectangle.
It contains a Primary key represented by an underline.	It contains a Partial Key or discriminator represented by a dashed underline.
The member of strong entity set is called as dominant entity set.	The member of weak entity set is called as subordinate entity set.
The Primary Key is one of its attributes which uniquely identifies its member.	The Primary Key of weak entity set is a combination of partial key and primary key of the strong entity set.
The relationship between two strong entity set is represented by a diamond symbol.	The relationship between one strong and a weak entity set is represented by a double diamond sign. It is known as Identifying relationship.
The line connecting strong entity set with the relationship is single.	The line connecting weak entity set with the identifying relationship is double.
Total participation in the relationship may or may not exist.	Total participation in the identifying relationship always exists.

Attributes

- Entities are represented by means of their properties, called **attributes**.
- In other words, **attributes** are the **properties / characteristics** of an entity.
- All attributes have values.
- For example, a student entity may have
 - **rollno, name, class, age** as attributes.
- There exist a domain or range of values that can be assigned to attributes.
- For example, a student's name cannot be a numeric value. It has to be alphabetic. A student's age cannot be negative, etc.

Types of Attributes

- Simple and Composite Attribute
- Single Valued and Multi Valued attribute
- Stored and Derived Attributes
- Complex Attribute

Simple and Composite Attribute

Simple attribute:

- Simple attribute consists of a single atomic value.
- It cannot be subdivided. For example, student's phone-number is an atomic value of 10 digits. The attributes **age, sex, salary** etc. are simple attributes.

Composite attribute:

- Composite Attribute value is non atomic.
- It is an attribute that can be further subdivided. Composite attributes are made of more than one simple attribute. For example:
- The attribute ADDRESS can be subdivided into **street, city, state, and pin code.**
- The attribute Name can be subdivided into **First Name : Middle Name : Last Name**

Single Valued and Multi Valued attribute

Single-valued attribute:

- A single valued attribute can have only a single value.
- For example a person can have only one **“date of birth”, “age”, “PAN No”, “Aadhar No.”** etc.
- But it can be simple or composite attribute. That is 'date of birth' is a composite attribute, 'age' is a simple attribute. But both are single valued attributes.

Multi-valued attribute:

- Multivalued attributes can have multiple values.
- For example a person may have **multiple phone numbers, multiple degrees, email addresses** etc.
- Multivalued attributes are shown by a double line connecting to the entity in the ER diagram.

Stored and Derived attributes

Complex Attribute

- **Stored Attribute**: An attribute that supplies a value to the related attribute.
 - Example: **"Date of Birth"**
- **Derived Attribute**: An attribute that's value is derived from a stored attribute.
 - Example : **"age"**, and it's value is derived from the stored attribute "Date of Birth".
 - The value for the attribute **'age'** can be derived by subtracting the **'Date of Birth'(DOB)** from the current date.
- **Complex Attribute**: A complex attribute that is both composite and multi valued.
 - Example: A person can have many **phone nos., e-mail ids**, etc. So, if **"Contact details"** contains **phone nos., e-mail ids**, then it will be a complex attribute.

E R Diagram

ER-Diagram is a visual representation of data that describes how data is related to each other.

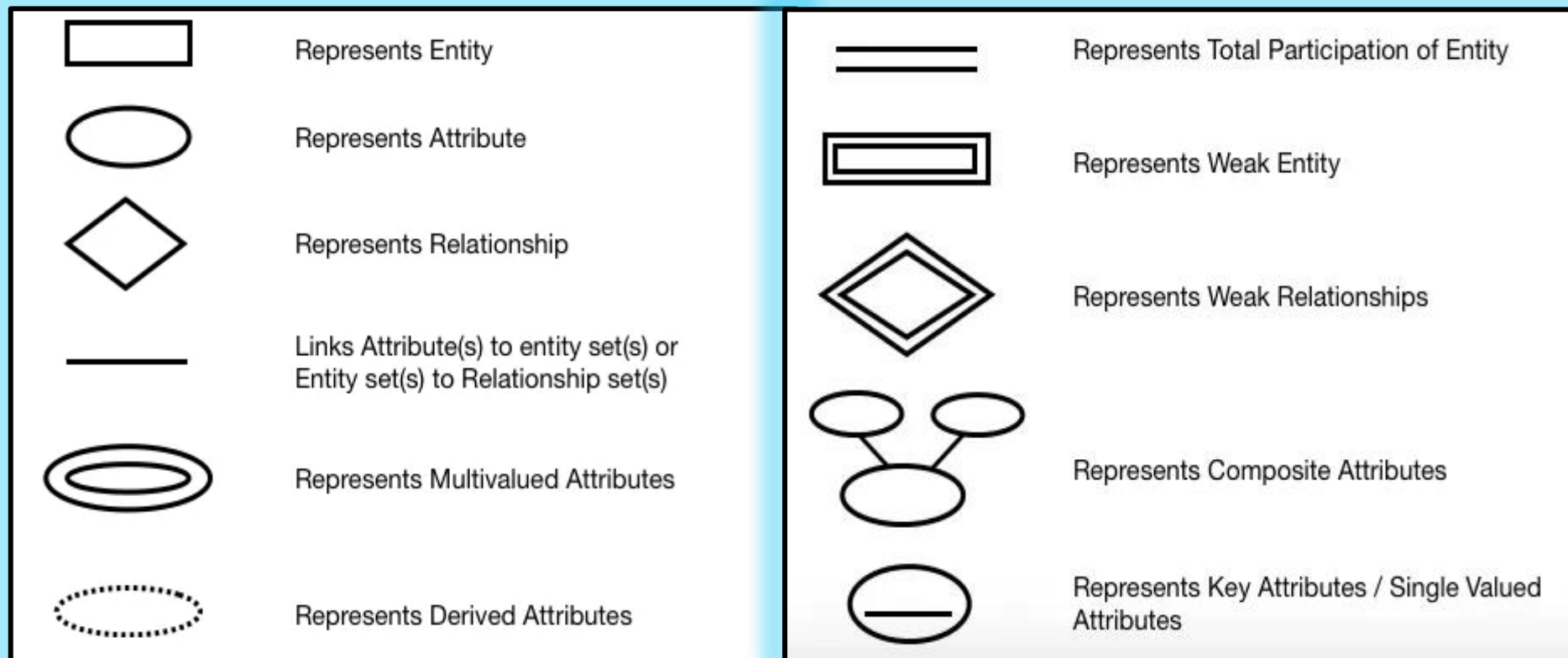
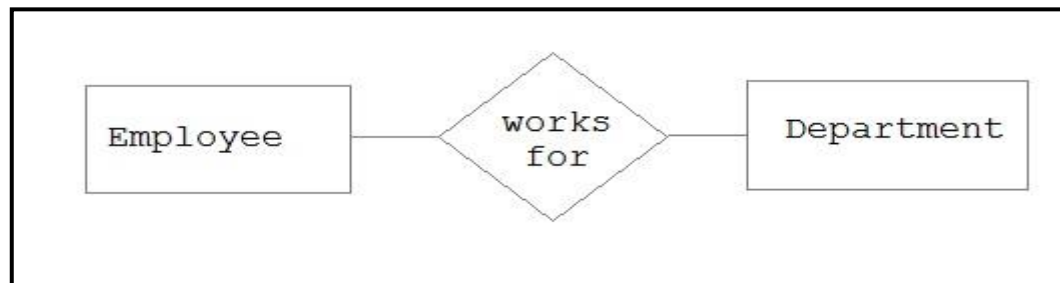
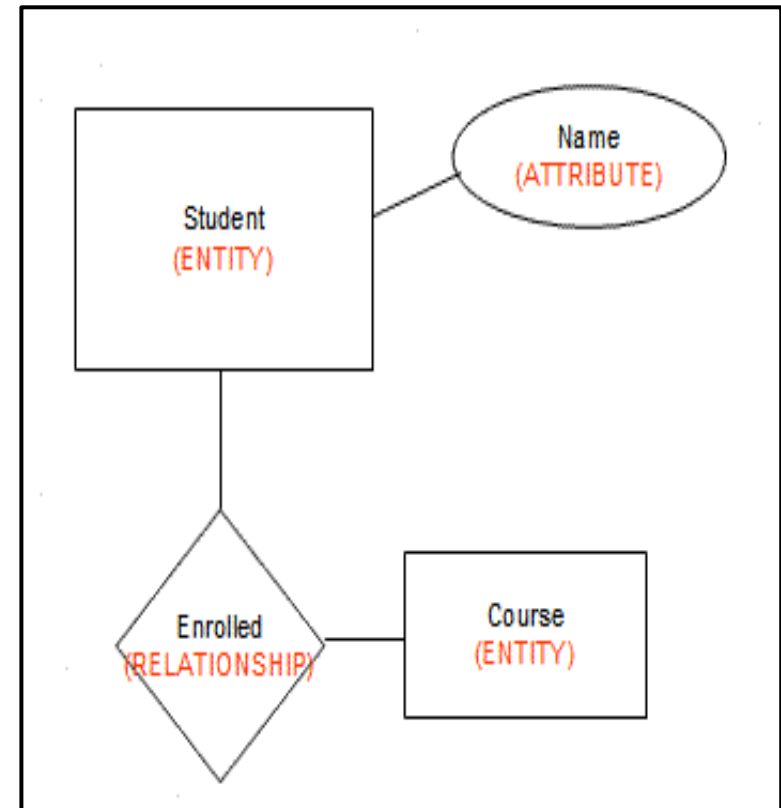
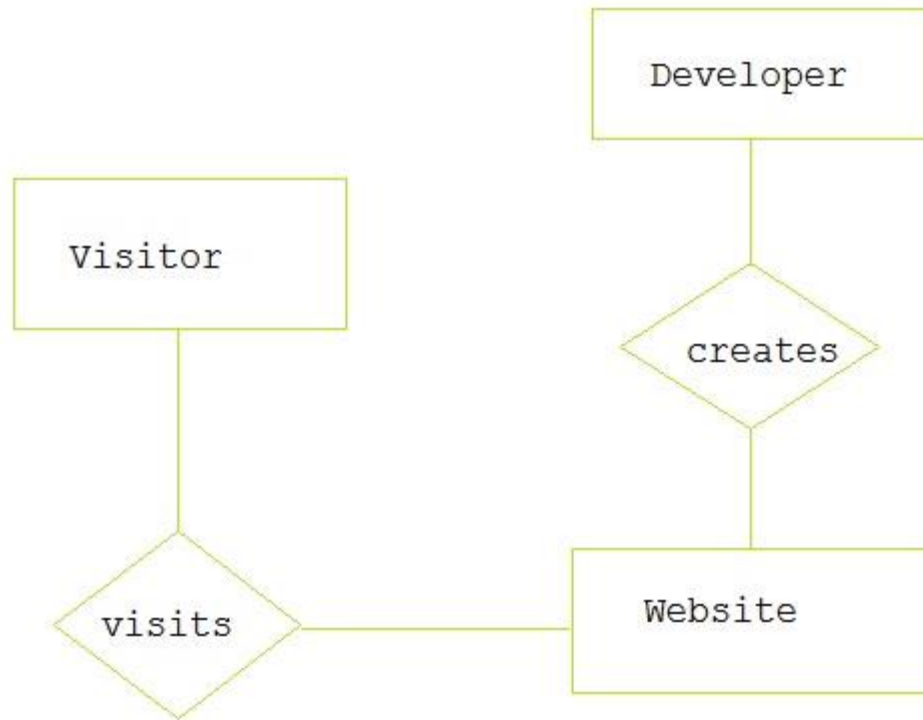
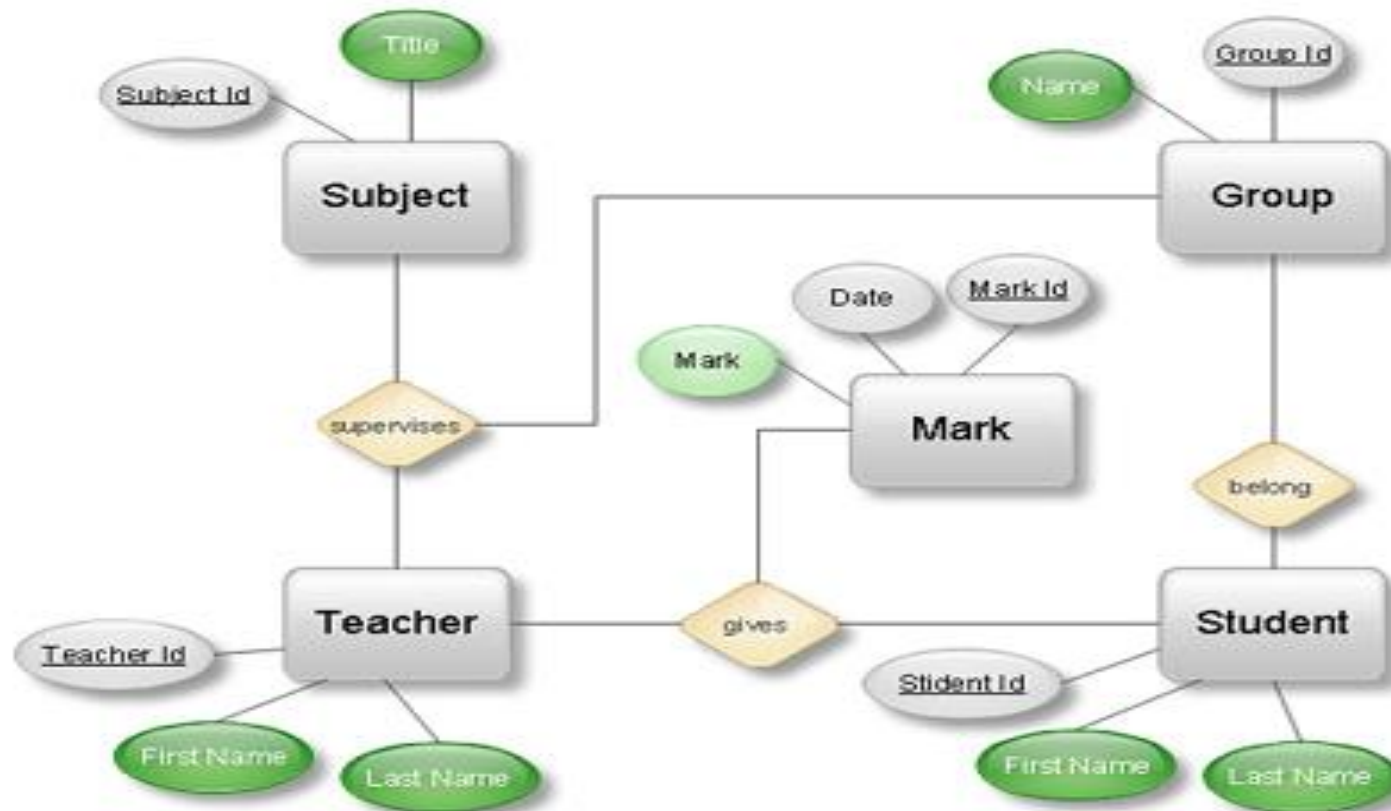


Figure : Symbols used in E-R Diagram(ERD)

Example : E R Diagram



Example : E R Diagram



Relationship

- The association among entities is called **relationship**.
- For example, **employee entity** has relation **works** for **department**.
- Another example is for **student** who **enrols** in some **course**.
- Here, **works** and **enrols** are called **relationship**.

Relationship Set:

- Relationship of similar type is called relationship set.
- Like entities, a relationship too can have attributes.
- These attributes are called descriptive attributes.

Degree of relationship

- The number of participating entities in a relationship defines the degree of the relationship.
 - **Binary Relationship = degree 2**
 - **Ternary Relationship = degree 3**
 - **Recursive Relationship**
 - **n-ary = degree**

Relationship

Binary Relationship

- Binary Relationship means relation between two Entities. This is further divided into Four types.
 - **One-to-One**
 - **One-to-Many**
 - **Many-to-One**
 - **Many-to-Many**

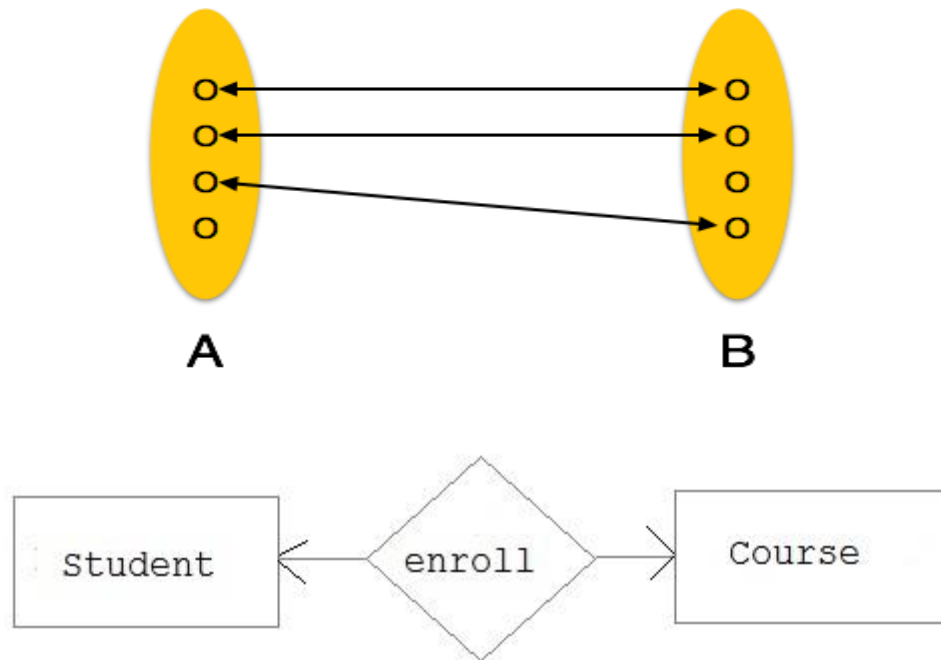
Mapping Cardinalities:

- **Cardinality** defines the number of entities in one entity set which can be associated to the number of entities of other set via relationship set.

Relationship

One-to-one: one entity from entity set A can be associated with at most one entity of entity set B and vice versa.

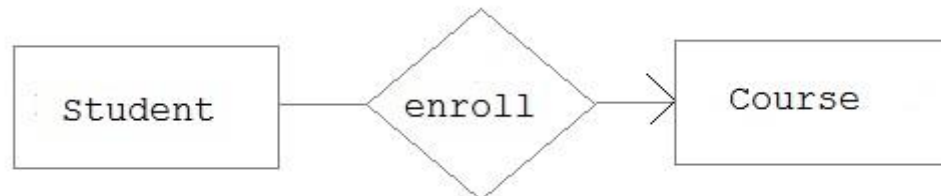
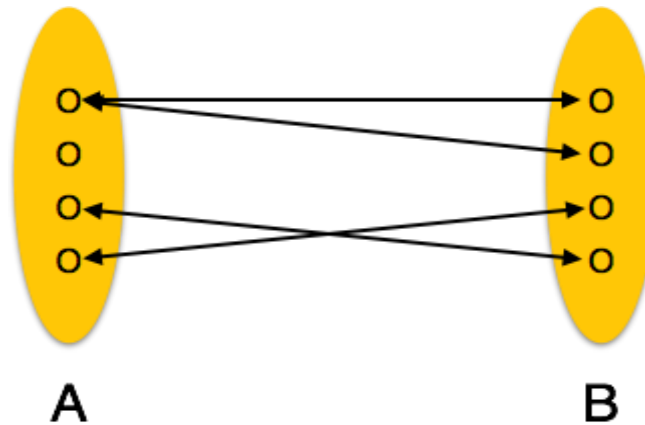
This type of relationship is rarely seen in real world.



The above example describes that one student can enrol only for one course and a course will also have only one Student. This is not what you will usually see in relationship.

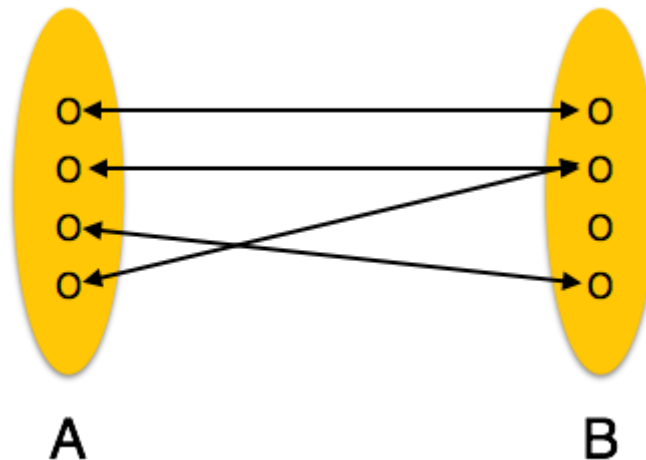
Relationship

One-to-many: One entity from entity set A can be associated with more than one entities of entity set B but from entity set B one entity can be associated with at most one entity.



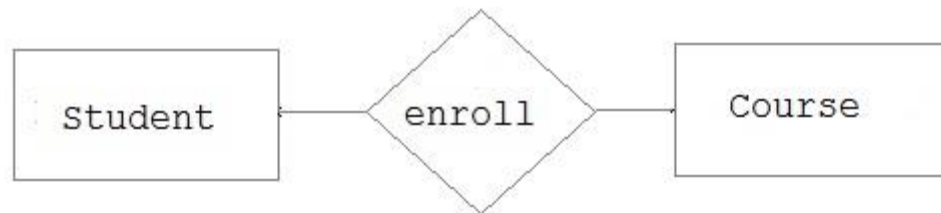
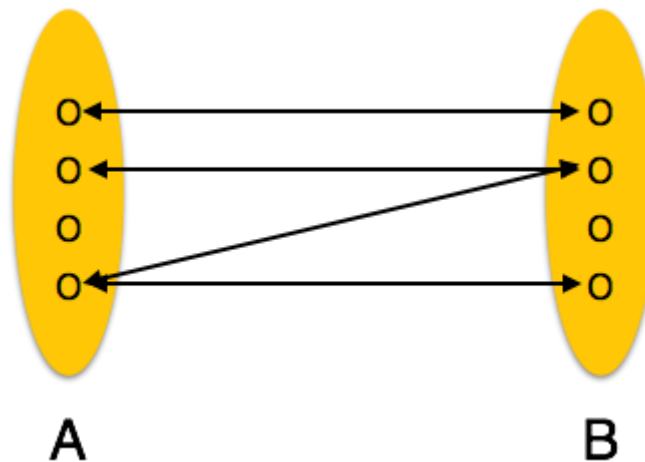
Relationship

Many-to-one: More than one entities from entity set A can be associated with at most one entity of entity set B but one entity from entity set B can be associated with more than one entity from entity set A.



Relationship

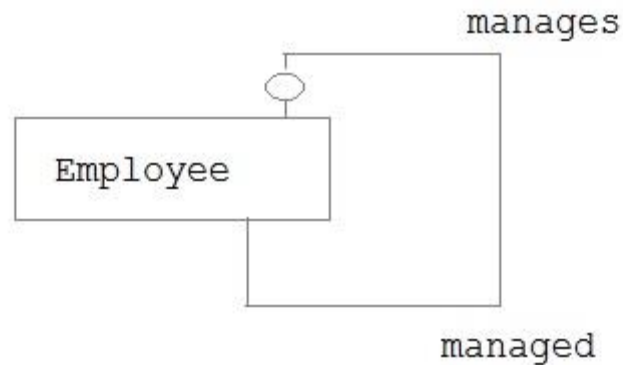
Many-to-many: one entity from A can be associated with more than one entity from B and vice versa.



Relationship

Recursive Relationship

- When an Entity is related with itself it is known as **Recursive** Relationship.



Thanks

Finally, We are able to draw an E-R
Diagram for a given situation.

